

PATCHWISE AGENT2AGENT (A2A) REVIEW - ONE PAGE

1) AIM

PatchWise A2A converts patch review from manual babysitting into autonomous, evidence-driven rounds.

It improves consistency, preserves context, and blocks unsafe LGTM decisions.

2) CORE IDEA

- Chanakya (Builder): proposes/fixes patch changes with explicit evidence and residual risk notes.
- Aryabhatta (Reviewer): performs adversarial review and emits structured findings.
- Orchestrator (A2A CLI): runs rounds, enforces gates, tracks findings lifecycle, writes reports.

3) BUILDING BLOCKS

- Session manager: .a2a/sessions/<session-id>.json
- Findings engine: open/new/resolved/closed tracking
- Validation gates: schema + optional checkpatch/static checks
- Lore intake: lore URL/msgid fetch + prior comment mapping
- Worktree lock: concurrent session safety
- Report engine: markdown/json/html outputs per round/session

4) CALL FLOW

Input (local patch / lore URL / msgid / resume session)

- > Session init
- > Builder round
- > Validation gate
- > Reviewer round
- > LGTM gate
 - > Pass: LGTM + reports + optional respin (vN -> vN+1)
 - > Fail: findings injected into next round and continue
- > Stop rules: max rounds/time budget/operator decision

5) TASKS COVERED

- Local patch review loop
- Lore-based patch series review
- Prior-thread comment mapping and suggested replies
- Multi-round fix-and-recheck
- Resume interrupted sessions
- Auto-generate next version patchset after LGTM

6) ACHIEVABLE OUTCOMES

- Reduced manual supervision effort
- Better reviewer consistency and traceability
- Faster iteration with preserved history
- Deterministic evidence bundle for maintainers

7) ESTIMATED TIME SAVING (INDICATIVE)

- 3-round patch cycle: Manual 120-180 min -> A2A 40-70 min (45-65% saving)
- Context rebuild after interruption: Manual 20-40 min -> A2A 3-8 min (70-90% saving)
- Report/evidence packaging: Manual 25-45 min -> A2A 5-12 min (65-85% saving)

8) POSSIBLE USE CASES

- Kernel patch pre-submit quality gate
- Lore review triage and response preparation
- Team handoff using session-id resume
- Remote review control with email bridge
- Version-to-version patchset coherence checks

9) COMMAND EXAMPLES

- a2a loop --task <name> --watch-path /abs/path --max-rounds 3
- a2a loop --task <name> --lore-url "https://lore.kernel.org/all/<msgid>/" --max-rounds 3
- a2a loop --session <session-id>
- ./run.sh <lore-url|msgid|session|path>

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